SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

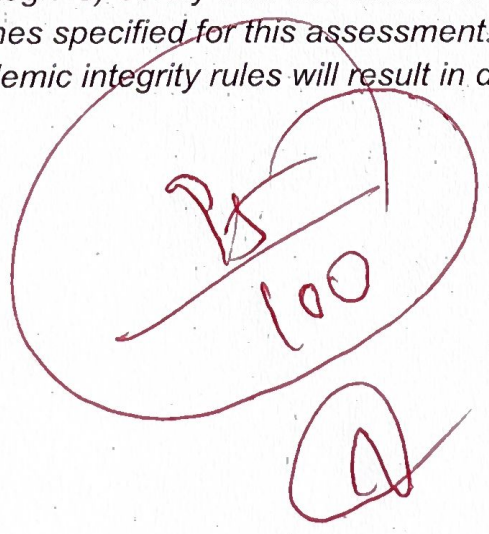
Annexure A

CONCEPT MAPPING

Code of Conduct:

I V.Pradhyumna(192425336) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
    a. Show how each attack leads to specific threats and risks
    b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
    a. Show differences and similarities
        i. Map how each affects the Data structure and Security level
        ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
    a. Show relationships between these concepts
    b. Analyze how each contributes to secure communication
    c. Include one real-world application for each

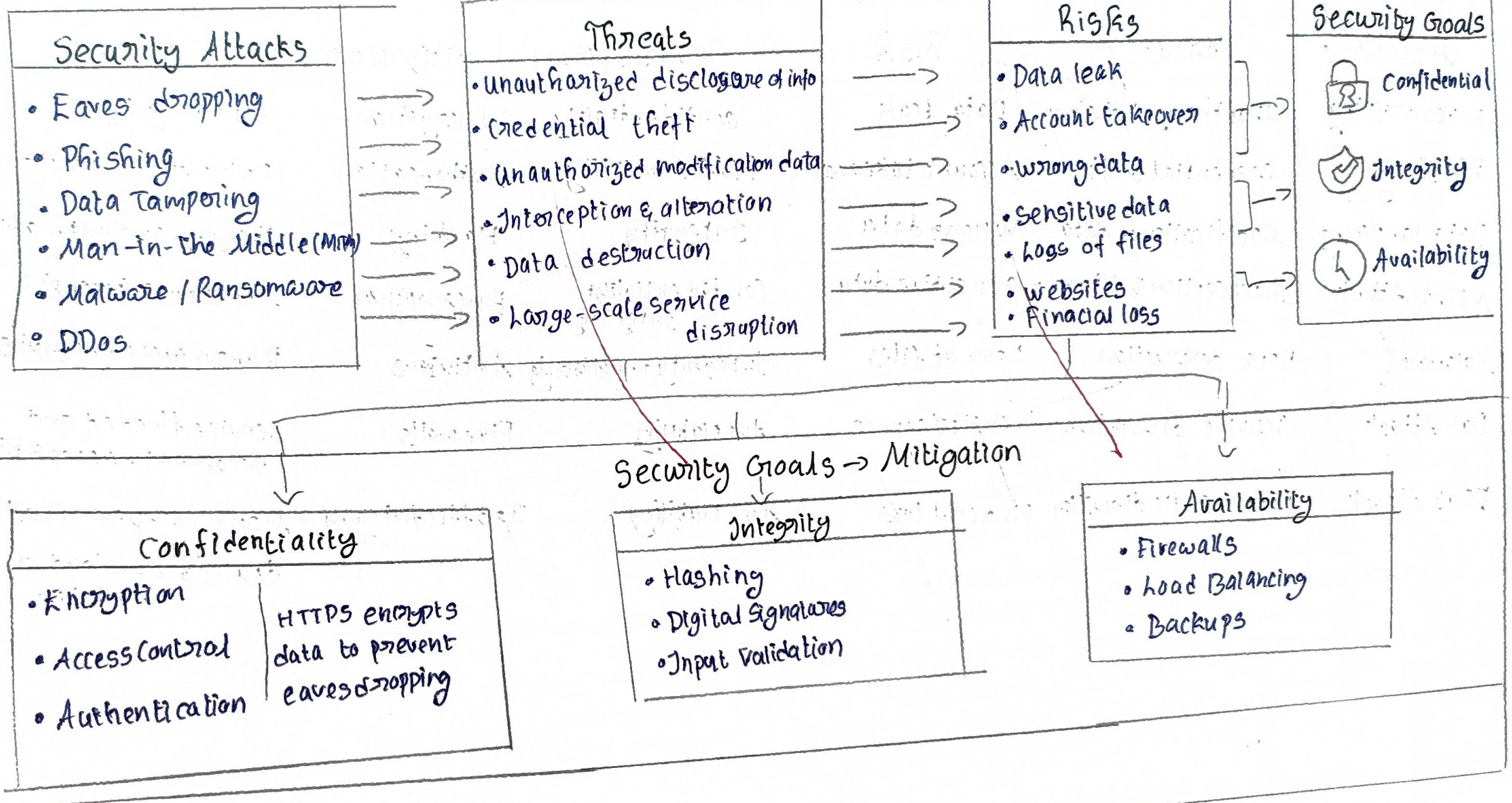
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
    a. Show how each concept is mathematically connected
    b. Map their role in encryption/decryption
    c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

①

Concept Map : Security Attacks, Threats, Risks and Security Goals



Real-World Example for Each link

Attack	Threat	Risk	Security Goal	Mitigation	Example.
Eavesdropping	Unauthorized disclosure	Data leak	confidentiality	Encryption	capturing login
Phishing.	Credential theft	Account takeover	Confidentiality	Authencation	Fake email tricks
Data Tampering	Modification of data	Wrong data	Integrity	Hashing	changing amount in online
MITM Attack	Interception & Alteration	Sensitive data	Confidentiality Integrity	Encryption	Attacker intercepts
Malware	Data destruction	Loss of files	Integrity + Availability	Antivirus	Ransomware encrypts
Dos Attack	Service disruption	Website/server	Availability	Firewalls	service flooded and becomes inaccessible
DDos Attack	Large-scale disruption	Financial loss	Availability	Traffic filtering	Many compromised devices overload a website.

(2)

Substitution Techniques in Cryptography

Caesar Cipher
Each letter is shifted by fixed no. of positions in the alphabet

Monoalphabetic cipher
Each letter is substituted with a fixed, unique letter using a substitution alphabet

Vigenere cipher
Uses a keyword to apply different shifts for different positions

Hill cipher
Uses a matrix key to encrypt a block of letters at a time using linear Algebra

	Caesar Cipher	Monoalphabetic cipher	Vigenere cipher	Hill cipher
Key usage	single numerical key	one substitution alphabet	Keyword	matrix key
Complexity	very low	low	Medium	High
Security Strength	very weak	weak to moderate	strong	stronger

Analysis

Caesar cipher
- Simplest technique
- Easy to implement
- Small key space
- Can be easily broken using brute force

Monalphabetic Cipher
- Uses a fixed substitution
- More complex than caesar
- Vulnerable to frequency
- Provides low to moderate security

Vigenere cipher
- Uses a keyword
- Multiple shifting alphabets
- Stronger than caesar and monalphabetic
- Can be broken if key is a short

Hill cipher
- Uses matrix operations
- Encrypts multiple letters together
- High on mathematical complexity
- Stronger diffusion